Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

ISO 14064-1:2018 Greenhouse Gas (GHG) Emissions and Removals

TGS Ref No: TGS-2025055502

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	Legacy release	Legacy trainer slide deck.	Tertiary Infotech Academy
2.0	20 August 2026	Complete source-aligned visual rebuild; expanded ISO 14064-1 inventory, risk, reduction, renewable-energy and offset content; ten connected activities and aligned assessments.	Dr Alfred Ang

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Course Information

Course	ISO 14064-1:2018 Greenhouse Gas (GHG) Emissions and Removals
Course Code	TGS-2025055502
TSC	Greenhouse Gas Accounting · ACC-ADV-0002-1.1
Duration	2 days · 16 training hours
Daily timing	9:00 am-6:00 pm; 60-minute lunch; two 10-minute breaks within training time
Assessment	Day 2, 4:00-6:00 pm · WA 1 hour + Case Study 1 hour

Learning Outcomes

- LO1: Innovate organisational GHG accounting systems using current climate science, ISO 14064-1 principles and broader sustainability initiatives.
- LO2: Develop risk-informed GHG management plans that meet relevant reporting and regulatory requirements and support mitigation decisions.
- LO3: Develop operational emissions-reduction strategies across product lifecycles, value chains, supply chains and renewable-energy options.
- LO4: Generate economically viable carbon-offset and carbon-credit strategies aligned with organisational targets and residual-emissions needs.

Course Schedule

Day 1 — ISO 14064-1 inventory design, boundaries, quantification, reporting and risk controls

Time	Duration	Topic / Activity	Slide reference	Method
9:00-9:30	30 min	Digital attendance, course orientation, introductions, learning outcomes and assessment briefing	Slides 1-19	Facilitated briefing and learner introduction
9:30-10:30	60 min	GHG science, ISO 14064 family, principles, boundaries and inventory purpose	Slides 20-34	Interactive presentation and worked examples
10:30-10:40	10 min	Tea break	-	Break
10:40-11:40	60 min	Activity 1 — Set the GHG Inventory Purpose and Boundaries	Slides 35-37	Group case study and workbook
11:40-13:00	80 min	Activity 2 — Build a Significant GHG Source and Sink Register	Slides 38-40	Source classification and significance workshop
13:00-14:00	60 min	Lunch break	-	Break

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Time	Duration	Topic / Activity	Slide reference	Method
14:00-15:15	75 min	Activity 3 — Quantify a Base-Year Inventory and Reporting Guide	Slides 41-44	Calculation, reconciliation and peer check
15:15-15:30	15 min	Reporting, inventory review, governance and GHG risk controls	Slides 45-57	Visual explanation and defect analysis
15:30-15:40	10 min	Tea break	-	Break
15:40-16:55	75 min	Activity 4 — Review a GHG Inventory and Resolve Quality Findings	Slides 58-60	Evidence review and finding calibration
16:55-17:45	50 min	Activity 5 — Build a GHG Governance, Compliance and Risk Plan	Slides 61-63	Risk, control and governance design
17:45-18:00	15 min	Day 1 recap, Q&A and PM digital attendance	Slides 64-65	Retrieval practice and reflection

Total training time: 480 minutes (8 hours).

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Day 2 — Value-chain reduction, renewable energy, offsets and integrated carbon strategy

Time	Duration	Topic / Activity	Slide reference	Method
9:00-9:15	15 min	Day 1 retrieval practice and AM digital attendance	Slide 65	Facilitated recap
9:15-10:05	50 min	Hotspots, lifecycle thinking, supplier data, targets and reduction hierarchy	Slides 66-78	Visual explanation and worked examples
10:05-10:30	25 min	Activity 6 — Diagnose Hotspots and Set a Reduction Baseline (part 1)	Slides 79-81	Hotspot and lifecycle analysis
10:30-10:40	10 min	Tea break	-	Break
10:40-11:05	25 min	Activity 6 — Diagnose Hotspots and Set a Reduction Baseline (part 2)	Slides 79-81	Opportunity prioritisation and acceptance check
11:05-12:05	60 min	Activity 7 — Design a Supplier and Product Reduction Roadmap	Slides 82-84	Supplier segmentation and roadmap design
12:05-13:00	55 min	Activity 8 — Develop a Renewable-Energy Business Case	Slides 85-88	Scenario modelling and recommendation
13:00-14:00	60 min	Lunch break	-	Break
14:00-14:30	30 min	Residual emissions, offset integrity, economics, claims and credit portfolios	Slides 89-101	Interactive presentation and decision cases
14:30-15:20	50 min	Activity 9 — Evaluate Carbon Offset Projects and Economics	Slides 102-104	Due diligence and risk-adjusted economics
15:20-15:30	10 min	Tea break	-	Break
15:30-16:00	30 min	Activity 10 — Integrate the Organisational Carbon Strategy	Slides 105-108	Capstone strategy and board decision
16:00-17:00	**60 min**	**Written Assessment (WA-SAQ): 9 questions covering K1-K9**	**Slides 109-114**	**Individual, summative, open book**
17:00-18:00	**60 min**	**Case Study Assessment: 4 questions covering A1-A8; assessment attendance and submission**	**Slides 114-116**	**Individual, summative, open book**

Total training time: 480 minutes (8 hours).

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Activity and Competency Map

Activity	Title	Alignment	Evidence/output
1	Set the GHG Inventory Purpose and Boundaries	K1 · K2 · K5 · A3 · LO1	Inventory purpose statement; Organisational boundary diagram; Consolidation decision log; Peer-review record
2	Build a Significant GHG Source and Sink Register	K1 · K5 · A3 · A7 · LO1	Completed source/sink register; Significance criteria and scores; Inclusion/exclusion log; Data-owner matrix
3	Quantify a Base-Year Inventory and Reporting Guide	K1 · K2 · K5 · A3 · A7 · LO1	Quantified inventory workbook; Factor and GWP register; Uncertainty improvement log; Base-year and reporting guideline
4	Review a GHG Inventory and Resolve Quality Findings	K5 · K6 · A4 · LO2	Review plan and sample log; Finding register; Corrected report disclosures; Management approval brief
5	Build a GHG Governance, Compliance and Risk Plan	K6 · K7 · A4 · A8 · LO2	Compliance register; GHG governance RACI; Risk and control register; Monitoring dashboard and escalation protocol
6	Diagnose Hotspots and Set a Reduction Baseline	K3 · A1 · LO3	Hotspot dashboard; Lifecycle map; Baseline and intensity statement; Prioritised opportunity shortlist
7	Design a Supplier and Product Reduction Roadmap	K3 · K9 · A1 · LO3	Supplier segmentation and data ladder; Quantified initiative cards; 24-month roadmap; KPI and governance plan
8	Develop a Renewable-Energy Business Case	K4 · A6 · LO3	Renewable-energy scenario model; Option scorecard; Sensitivity analysis; Executive recommendation
9	Evaluate Carbon Offset Projects and Economics	K8 · K9 · A2 · LO4	Due-diligence register; Quality scorecard; Risk-adjusted cost model; Procurement and claims recommendation
10	Integrate the Organisational Carbon Strategy	K2 · K7 · K8 · K9 · A2 · A5 · LO4	2030 pathway model; Prioritised initiative portfolio; Offset and credit policy; Board dashboard and decision memo

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